

C4

Project Climate Conditions

New York City falls within ASHRAE Climate Zone 4A (mixed-humid), characterized by hot, humid summers and cold winters. Average high temperatures reach approximately 84°F (29°C) in July and drop to about 38°F (3°C) in January. Annual precipitation is approximately 50 inches, distributed relatively evenly throughout the year, with snowfall averaging 25–30 inches. Solar exposure at the site is significantly affected by surrounding supertall towers, which cast long shadows particularly during winter months. Prevailing winds are from the northwest in winter dramatically altered by the tower cluster. The faceted envelope creates variable solar exposure on each angled surface, enabling surface-specific strategies for glazing, insulation, and shading. The waterfront proximity moderates temperatures slightly but increases humidity and wind exposure.

Key Considerations

- ASHRAE Zone 4A (mixed-humid): hot summers, cold winters, ~50 in annual precipitation
- Summer avg high: 84°F (29°C); winter avg high: 38°F (3°C); snowfall 25–30 in
- Solar access reduced by supertall tower shadows, especially in winter months
- Prevailing winds NW (winter), SW (summer), locally altered by tower canyon effects
- Faceted envelope creates variable solar exposure, enables surface-specific strategies

